

Cancer Surveillance: Handout

How Cancer Data Becomes Public Health Action

CPH 705 - Asynchronous Session Handout

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Cancer surveillance tracks who gets cancer, who dies from it, and how patterns change over time. It turns everyday clinic and hospital visits into “big picture” information that guides screening, resource planning, and equity-focused interventions.

1. What does cancer surveillance do?

- Ongoing, systematic collection, analysis, and sharing of cancer data.
- Describes how common different cancers are (burden).
- Tracks trends over time (getting better or worse).
- Identifies who is most affected (age, sex, race/ethnicity, geography).
- Breast, lung, and prostate cancers are among the most commonly diagnosed worldwide, and lung cancer remains the leading cause of cancer death.

2. How are cancer data collected?

- **Passive:** Facilities send reports to a central registry.
 - Pros: Efficient for routine, large-scale monitoring.
 - Cons: Some cases may be missed.
- **Active:** Registry staff search for and verify cases.
 - Pros: More complete and accurate.
 - Cons: Requires more time and staff.
- **Sentinel:** Selected expert sites report in detail (e.g., rare cancers).
 - Pros: Detailed, high-quality information.
 - Cons: Not fully representative.
- **Syndromic:** Uses near-real-time signals (ED visits, pharmacy data).

- Pros: Early warning for emerging issues.
- Cons: Non-specific and needs confirmation.

3. Turning charts into comparable data

- **ICD-O-3 coding:**

- Topography = where the tumor started.
- Morphology = what the cells look like and how they behave.
- Behavior codes: /3 malignant (reportable); /2 in situ (usually reportable); /0 benign (usually not, except some CNS tumors).

- **Staging:**

- AJCC TNM: detailed staging for individual care.
- Summary Stage (Localized / Regional / Distant): simpler categories for population trends.

4. Data quality and "spikes"

- Key ideas: completeness, accuracy, timeliness, and consistency.
- A sudden increase after an EHR change may reflect better case finding (a completeness artifact), not a true jump in cancer risk.